

Markov Chains



Markov Chains



Suppose there are only two possible kinds of weather: sunny and rainy.

Tomorrow's weather is random, but we might expect it to depend on today's weather. If today is sunny, perhaps tomorrow is more likely to be sunny than rainy.

A Markov chain makes one additional assumption. Once we know what the weather is today, knowing what happened yesterday or two days ago does not give us any additional information about tomorrow.

So the main idea of a Markov chain is that the future **depends on the present, but not directly on the entire past**.

Markov Chains



Markov chains are useful because many random systems don't just produce one random outcome. Instead, the system has a **state**, and that state changes over time.

For example,

- In a weather model, the state might be sunny or rainy.
- In a board game, the state might be the square that a player currently occupies.
- When someone is browsing the internet, the state could be the webpage they're currently visiting.
- In a queueing system, the state might be the number of customers currently waiting.

In each of these examples, the next state is not necessarily deterministic. Instead, there are different possible next states, each with some probability.

Markov Chains

Let X_1, X_2 be the state of the system over time.

$$P(X_{n+1} = j \mid X_n = i, X_{n-1}, \dots, X_0) = P(X_{n+1} = j \mid X_n = i).$$

Given the present, the future is independent of the past.

Let X_n represent the state of the system at time n .

The Markov property says that the probability of the next state, X_{n+1} , given the entire history of the process, is the same as the probability of the next state given only the current state X_n .

In other words, once we know where we are now, knowing how we got here doesn't help us predict the next step.

The Markov property does **not** say that all of the states are independent.

For example, tomorrow's weather can still depend very strongly on today's weather. If it's sunny today, that might make sunny weather tomorrow much more likely.

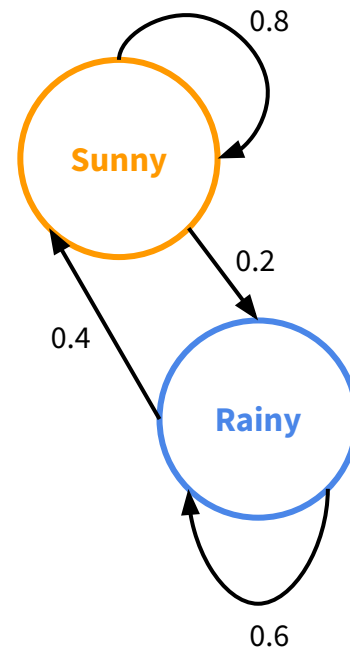
A Weather Markov Chain

$$S=\{\text{Sunny},\text{Rainy}\}$$

Suppose that if today is sunny, tomorrow will also be sunny with probability 0.8. Therefore, tomorrow is rainy with probability 0.2.

If today is rainy, suppose tomorrow becomes sunny with probability 0.4, and it remains rainy with probability 0.6.

We can represent these probabilities using a diagram like this, but there is an even more useful representation: a matrix.



Transition Matrix

Rows = current state

Columns = next state

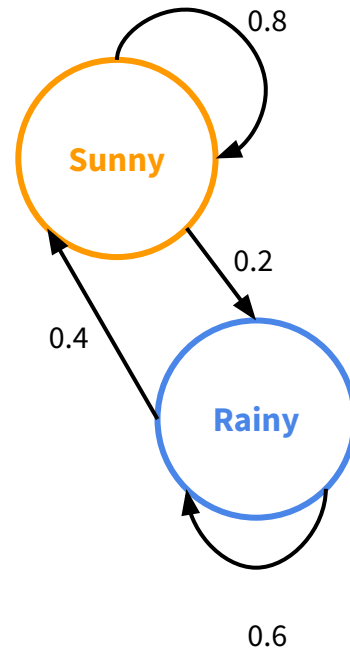
$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$

Each row sums to 1.

So 0.8 is the probability of going from Sunny to Sunny, and 0.2 is the probability of going from Sunny to Rainy.

The second row represents starting in the Rainy state. There is a 0.4 probability of becoming Sunny and a 0.6 probability of staying Rainy.

The transition matrix is especially useful because matrix multiplication lets us calculate what happens after more than one step.



Transition Matrix

Suppose today is Sunny.

$$\pi_0 = (1, 0)$$

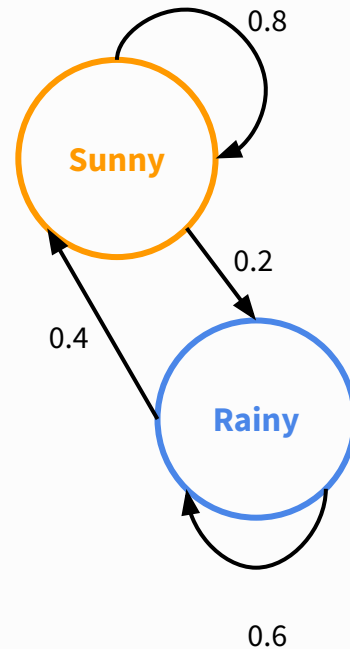
Tomorrow:

$$\pi_1 = \pi_0 P$$

$$(1, 0) \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = (0.8, 0.2)$$

So:

$$P(S) = 0.8, \quad P(R) = 0.2.$$



Transition Matrix

Two days from now:

$$\pi_2 = \pi_0 P^2$$

$$P^2 = \begin{pmatrix} 0.72 & 0.28 \\ 0.56 & 0.44 \end{pmatrix}$$

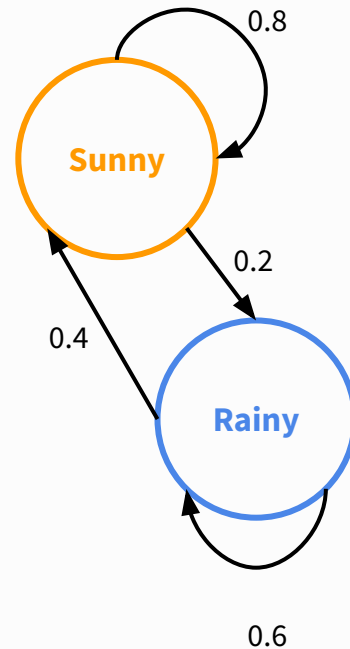
Therefore,

$$\pi_2 = (0.72, 0.28).$$

Why 0.72?

$$0.8(0.8) + 0.2(0.4) = 0.72$$

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$



Transition Matrix

Starting from Sunny:

$$\pi_0 = (1, 0)$$

$$\pi_1 = (0.8, 0.2)$$

$$\pi_2 = (0.72, 0.28)$$

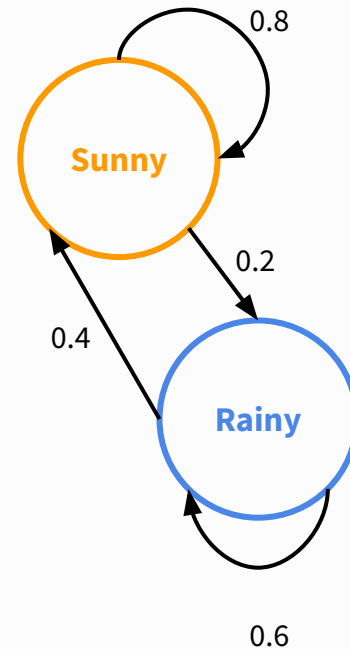
$$\pi_3 = (0.688, 0.312)$$

$$\pi_4 = (0.6752, 0.3248)$$

Eventually:

$$\pi_n \rightarrow (0.667, 0.333)$$

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$



Stationary Distribution

A stationary distribution π satisfies

$$\pi P = \pi$$

For our example, let

$$\pi = (s, r).$$

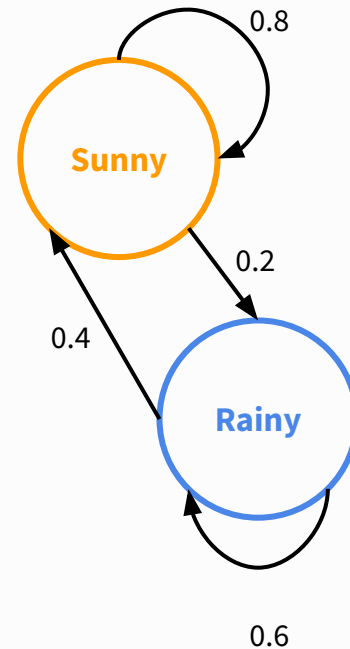
Then

$$(s, r) \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix} = (s, r).$$

Also:

$$s + r = 1.$$

$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$



Stationary Distribution

$$\pi = \left(\frac{2}{3}, \frac{1}{3} \right)$$

Long-run interpretation

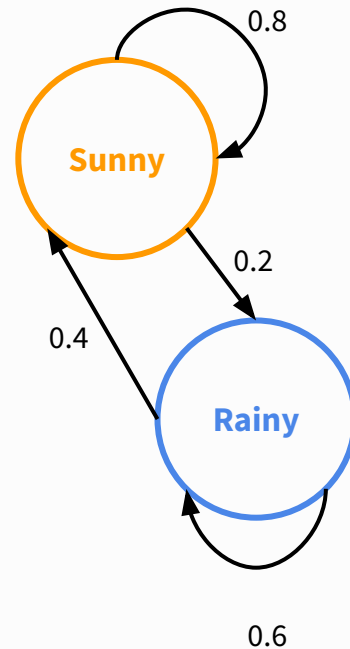
$$P(\text{Sunny}) \approx \frac{2}{3}$$

$$P(\text{Rainy}) \approx \frac{1}{3}$$

But...

Not every Markov chain converges.

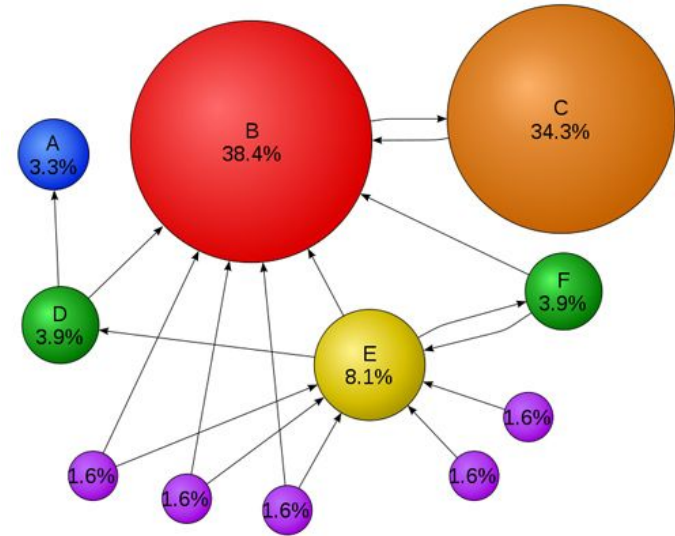
$$P = \begin{pmatrix} 0.8 & 0.2 \\ 0.4 & 0.6 \end{pmatrix}$$



Applications of Markov Chains

Although our weather example is very simple, the same mathematical idea appears in much more complicated systems.

One famous example is **PageRank**, the algorithm originally associated with ranking webpages for Google. We can imagine a random user moving from one webpage to another by following links. The webpages are states, and clicking links creates transitions between them.



<https://commons.wikimedia.org/w/index.php?curid=3470389>

Summary

1. Markov property

$$P(\text{future} \mid \text{present, past}) = P(\text{future} \mid \text{present})$$

2. Transition matrix

$$P_{ij} = P(X_{n+1} = j \mid X_n = i)$$

3. Multiple steps

$$P^n$$

4. Stationary distribution

$$\pi P = \pi$$

Sources

1. Norris, J. R. Markov Chains. Cambridge: Cambridge University Press, 1997.
2. Wikipedia contributors. “Markov Chain.” Wikipedia, The Free Encyclopedia.
https://en.wikipedia.org/wiki/Markov_chain.